Business Insights Report: E-commerce Customer Intelligence

Project: Customer Lifetime Value (CLV) & Profitability Command Center

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1. Executive Summary

This project involved the transformation of 10,000+ raw e-commerce transactions into a strategic decision-making tool. By implementing RFM (Recency, Frequency, Monetary) segmentation and CLV modeling, the analysis identifies $7.84M in total revenue and provides a roadmap for increasing profitability through targeted retention.

2. Methodology & Data Modeling

To move beyond basic reporting, three advanced analytical frameworks were built directly into the Excel engine:

- **CLV Modeling:** Used the FLOOR and AVERAGE functions to project the future value of customers based on historical lifespan and purchase frequency.
- **RFM Segmentation:** Customers were categorized into tiers (VIP, High, Mid, Low Value) based on their engagement patterns.
- **What-If Simulation:** A dynamic model was created to forecast how a 5% increase in retention or a 3% price adjustment would impact the bottom line.

3. Key Strategic Insights

- **The VIP Concentration:** While "VIP" customers make up a smaller portion of the total count (~601 customers), they maintain an average CLV of **$3,938**, nearly triple that of the "Mid Value" segment.
- **Churn Vulnerability:** Over **38% of the customer base** is currently flagged as "At Risk" (no purchase in 180+ days). This represents a significant revenue leakage that requires immediate re-engagement.
- **Profitability by Region: North America** and **Europe** lead in total volume, but **Asia** shows high efficiency in profit margins per order, suggesting lower relative logistics or marketing overhead.

- **Growth Trends:** Year-over-year analysis shows stable monthly revenue around **$330k - $350k**, indicating a mature market position with room for growth through "Whale" customer acquisition.

4. Data-Driven Recommendations

1. **Retention over Acquisition:** With an average CLV of $1,437, the cost to retain an existing High-Value customer is significantly lower than acquiring a new one. A loyalty program targeting "Mid Value" customers to move them to "High Value" should be prioritized.
2. **Regional Optimization:** Investigate the marketing strategies used in high-margin regions (Asia) and replicate those tactics in lower-margin territories to improve overall COGS efficiency.
3. **Predictive Intervention:** Use the **CLV Bin** data to automate discount triggers. When a customer with a high projected value enters the "At Risk" status, an automated win-back offer should be deployed.

5. Technical Specifications

The accompanying Excel workbook (CLV_Ecommerce_Dashboard.xlsx) features:

- **Dynamic Slicers:** Real-time filtering by Region, Segment, and Churn Status.
- **Pivot Backend:** A hidden processing layer that ensures the dashboard remains fast and responsive.
- **Automated Cleaning:** Formulas designed to handle raw CSV exports with minimal manual adjustment.